ECE 329 **Homework 5** **Due: Wed, Oct 29, 2025, 11:59 PM**

5 problems totaling 50pts.

1. (10pts) Verify that vector identity

$$\mathbf{H} \cdot \nabla \times \mathbf{E} - \mathbf{E} \cdot \nabla \times \mathbf{H} = \nabla \cdot (\mathbf{E} \times \mathbf{H})$$

holds for $\mathbf{E} = 2\hat{x}\cos(\beta z)$ and $\mathbf{H} = 4\hat{y}\cos(\beta z)$ by expanding both sides of the identity. Treat β as a real constant.

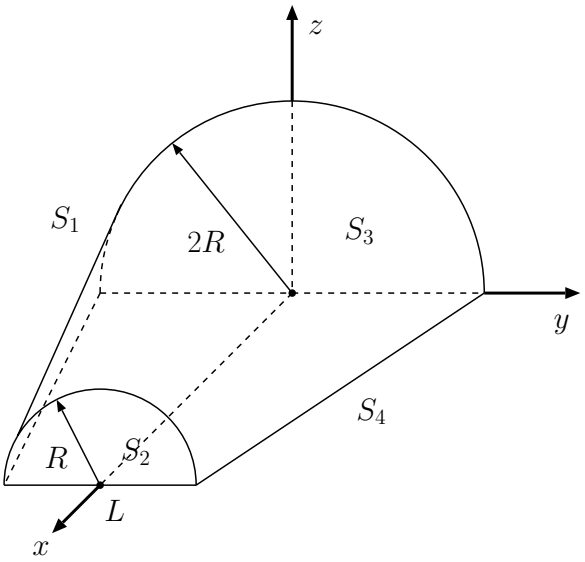
You should download the **table of vector identities** from ECE 329 web site and examine the list to familiarize yourself with the listed identities — they are widely employed in electromagnetics as well as in other branches of engineering such as fluid dynamics.

2. (10pts) In this problem we will compare the magnitudes of electrical force $q\mathbf{E}$ and magnetic force $q\mathbf{v} \times \mathbf{B}$ on a moving test charge q due to fields $\mathbf{E}$ and $\mathbf{B}$ produced by another charge Q at some distance r.

   a) (7pts) Let $\mathbf{E}$ be the static Coulomb field of Q with magnitude $\frac{Q}{4\pi\epsilon_o r^2}$. Also let $\mathbf{B}$ the magnetic field calculated using Biot-Savart law with a maximum magnitude $\frac{\mu_o I|d\mathbf{l}|}{4\pi r^2}$ where for $I|d\mathbf{l}|$ we can use QV_s, where V_s denotes the speed of charge Q when it is moving to produce $\mathbf{B}$. Denoting the speed of the test charge as v_q, show that the magnetic force on q can be as strong as the electric force only in the limit when the product of v_q and V_s approaches $c^2 \equiv \frac{1}{\mu_o\epsilon_o}$, where $c \approx 3 \times 10^8$ m/s. **Hint:** what is the expression for the largest possible ratio of the magnitudes of $q\mathbf{v_q} \times \mathbf{B}$ and $q\mathbf{E}$?

   b) (3pts) The result of part (a) shows that magnetic force is typically much weaker than the electrostatic force (when elementary sources are considered one at a time). Why then are magnetic effects observed in everyday life (e.g., pull and push of refrigerator magnets) often stronger than electrostatic effects (e.g., static fields that you experience while taking off your sweater)? Discuss.

3. (10pts) Gauss's law for magnetic field $\mathbf{B}$ states that the surface integral $\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$ over any closed surface S enclosing a volume V. Given that $\mathbf{B} = a\hat{x} + a\hat{y} - a\frac{\pi R}{L}\hat{z}$, determine the magnetic flux $\int \mathbf{B} \cdot d\mathbf{S}$ through the partial cone surface shown in the following figure:



4. (10pts) A pair of infinite current sheets located on $x = -1$ and $x = 1$ m surfaces carry the surface currents of $\mathbf{J}_s = 2\hat{y}$ A/m and $\mathbf{J}_s = -2\hat{y}$ A/m, respectively. Furthermore, an additional pair of infinite current sheets located on $y = 1$ and $y = -1$ m surfaces carry the surface currents of $\mathbf{J}_s = 2\hat{x}$ A/m and $\mathbf{J}_s = -2\hat{x}$ A/m, respectively.

   a) (1pt) Make a sketch of the described current configuration in terms of a 2D $x - y$ plot indicating the location of the current sheets and the flow directions of surface currents on these sheets.

   b) (9pts) Determine the magnetic field intensity $\mathbf{H} = \mathbf{B}/\mu_o$ in the following three regions (indicated in your diagram in part (a)):

      i. (3pts) $|x| < 1$ and $|y| < 1$.

      ii. (3pts) $x > 1$ and $|y| < 1$.

      iii. (3pts) $x > 1$ and $y > 1$.

5. (10pts) Consider an infinite slab (extending in x and z directions) of a finite width $W = 1$ m described by $|y| < W/2$. The slab is electrically neutral but it conducts a uniform current density of $\mathbf{J} = 2\hat{x}$ A/m^2 (meaning that it contains equal densities of positive and negative charge carriers moving in opposite directions parallel to $\hat{x}$). Outside the slab, that is for $y > W/2$ and $y < -W/2$, the charge and current densities are zero.

   a) (2pts) Using the right-hand-rule and Biot-Savart law discuss why the current slab should generate equal and opposite directed magnetic fields in $\pm\hat{z}$ directions in front of and behind the $y = 0$ plane.

   b) (2pts) Based on part (a), what is $\mathbf{B}$ on $y = 0$ plane?

   c) (2pts) Next, make use of the integral form of Ampere's law and the deductions of parts (a) and (b), to find $B_z(y)$ for $|y| > W/2$. Note that it should be possible to evaluate the line integral of $\mathbf{B} = B_z\hat{z}$ around a suitably defined rectangle just using simple multiplications.

   d) (2pts) Use Ampere's law to find B_z at a distance y within the current slab (i.e., $0 < y < W/2$).

   e) (2pts) Plot B_z as a function of y over $-W < y < W$.